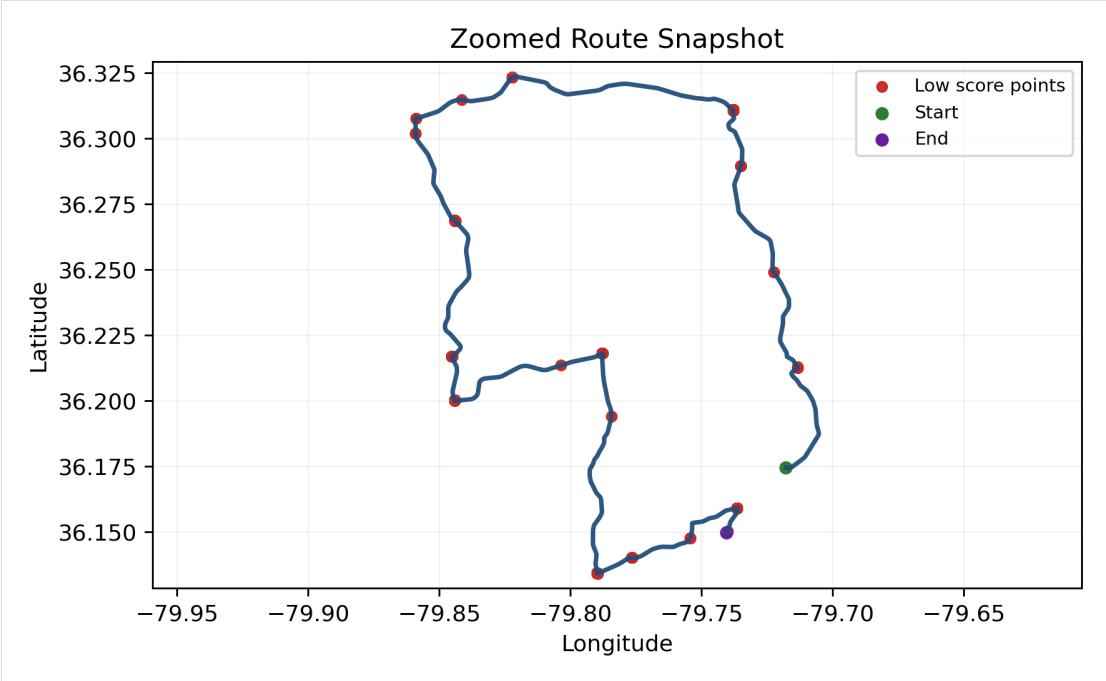
Rural Road Infrastructure Evaluation Report

Location

This experiment is conducted across Guilford County and Rockingham County in North Carolina.

U.S. State/County Context

Zoomed Route Snapshot



Evaluation Weights

These weights were read from the evaluated records used for this report. If the run was re-weighted, the values below reflect that updated configuration.

Weight	Value	Meaning
w1	0.6000	Physical infrastructure weight

Weight	Value	Meaning
w2	0.4000	Digital infrastructure weight
k1	1.0000	Lane-marking weight within physical infrastructure
k2	0.0000	Reserved physical component weight
k3	0.0000	Reserved physical component weight
m1	0.5000	Connectivity weight within digital infrastructure
m2	0.4000	GPS weight within digital infrastructure
m3	0.1000	HD-map weight within digital infrastructure

Executive Summary

Overall readiness for autonomous vehicle (AV) deployment on the surveyed rural road network is moderate-high: the dataset-level overall score mean is 0.8625 (std 0.0981), indicating generally favorable conditions for perception and localization but measurable deficits in communications and map availability.

Key facts and component-level evaluation - Dataset: 10,738 frame samples across 41 segments (segment snapshots provided for segments 0–7). - Overall score: mean = 0.862470 (std = 0.098127), min = 0.2225, max = 0.9475, median = 0.8725. - Lane detection (lane score): mean = 0.96907 (std = 0.16034). Distribution skewed to perfect scores (25/50/75% = 1.0). - Lane length pass rate: mean = 0.95081 (std = 0.17027). - Lane curvature pass rate: mean = 0.99283 (std = 0.08438). - Lane separation pass rate: mean = 0.96152 (std = 0.19232). - GPS (Global Positioning System) quality: mean = 0.98624 (std = 0.06238); median = 1.0, indicating consistently high localization accuracy. - Connectivity (5G / cellular link quality): mean = 0.61614 (std = 0.18143); quartiles: 25% = 0.4375, 50% = 0.5625, 75% = 0.75 — substantial variability and many low-quality frames. - HD maps (high-definition maps): mean = 0.0 (std = 0.0) — no HD map coverage in the measured frames.

Overall Performance

- Aggregate readiness: mean overall score 0.8625 (std 0.0981) — overall system/perception components perform well on average.
- Strong subsystems:
    - Perception / lane metrics: lane_score mean 0.9691 with median 1.0; curvature nearly always passes (mean curvature pass = 0.9928).
    - Localization: GPS mean 0.9862, low variance (std 0.0624) — localization is reliably high.
- Limiting subsystem:
    - Connectivity is the primary limiter (mean 0.6161, std 0.1814), with a broad interquartile spread indicating many frames with suboptimal network quality.
- HD maps are absent across all samples (hd_maps_score = 0.0), which removes a key redundancy for AV routing/validation.

Segment-level Issues

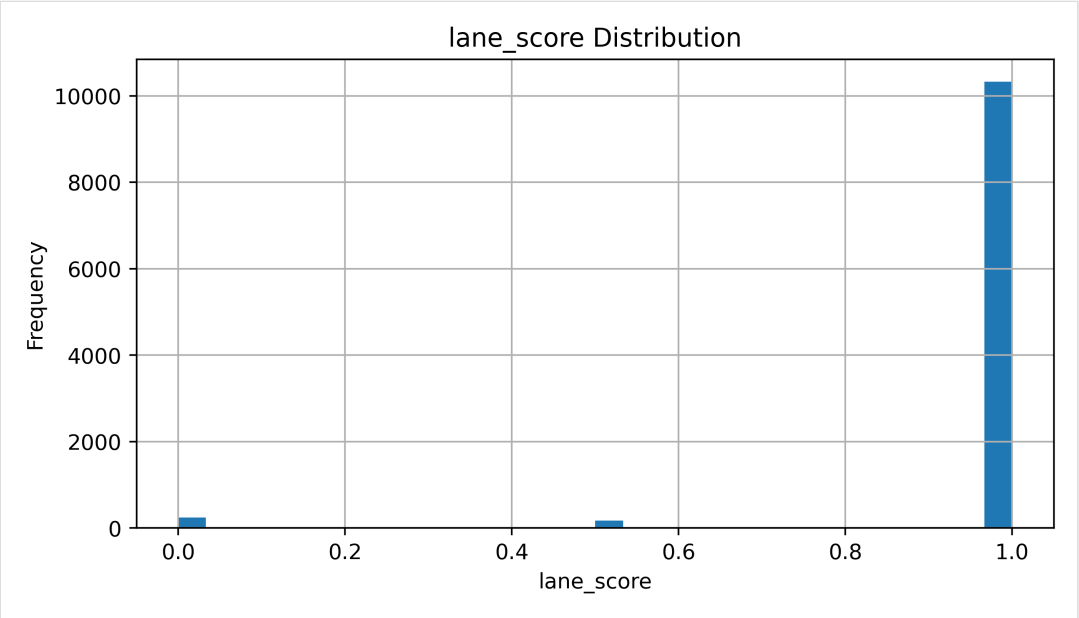
(Using provided segment snapshots 0–7; sample_count shown per segment) - HD-map coverage: - Every provided segment (0–7) reports "hd-map coverage low" for all samples in that segment (e.g., segment 0: 259/259 samples flagged; segment 3: 319/319). Global hd_maps_score = 0.0 confirms no HD map coverage in the dataset. - Connectivity problems (examples): - Segment 2: 97 connectivity-low occurrences out of 238 samples (40.8% of that segment). - Segment 1: 24 connectivity-low occurrences out of 263 samples (9.1%). - Segment 7: connectivity_score_avg = 0.5033 (mean) — below dataset mean and below median connectivity. - Connectivity averages vary by segment (examples: segment 1 = 0.7215, segment 2 = 0.5486, segment 7 = 0.5033). - GPS issues (rare): - Segment 0: 1 occurrence flagged "gps quality low" (1/259). - Lane-related weak checks (localized, low frequency): - Segment 1: "lane length check weak" flagged 2 times. - Segment 3: "lane quality low" flagged 5 times; "lane length check weak" 3 times. - Segment-level overall readiness lows: - Segment 3: 12 samples flagged "overall readiness low". - Segment overall score examples: - Highest among provided: segment 1 overall avg = 0.9013. - Lowest among provided: segment 7 overall avg = 0.8423. - Other examples: segment 2 = 0.8697, segment 3 = 0.8690.

Conclusions

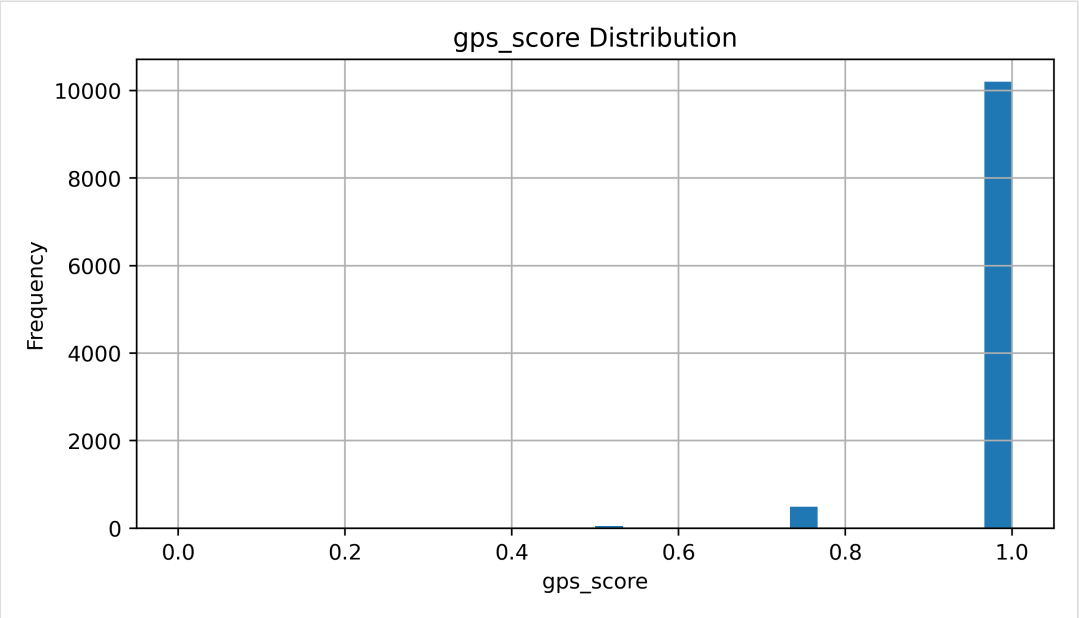
- Perception and localization readiness: High. Lane detection and GPS scores are both high (means 0.9691 and 0.9862 respectively; medians = 1.0), indicating these subsystems are largely suitable for AV tasks in the surveyed area.
- Primary readiness gaps:
    - Connectivity variability and low-quality frames (connectivity mean 0.6161, std 0.1814) represent the main operational risk for connected functions and remote services.
    - Absence of HD maps (hd_maps_score = 0.0) is a systemic gap across all sampled segments.
- Net implication: The dataset-wide overall score (~0.8625) reflects strong sensing/localization but diminished end-to-end readiness driven by connectivity and missing HD map coverage. Addressing those two factors would measurably raise segment and overall readiness (based on the observed score components and variances).

Figures

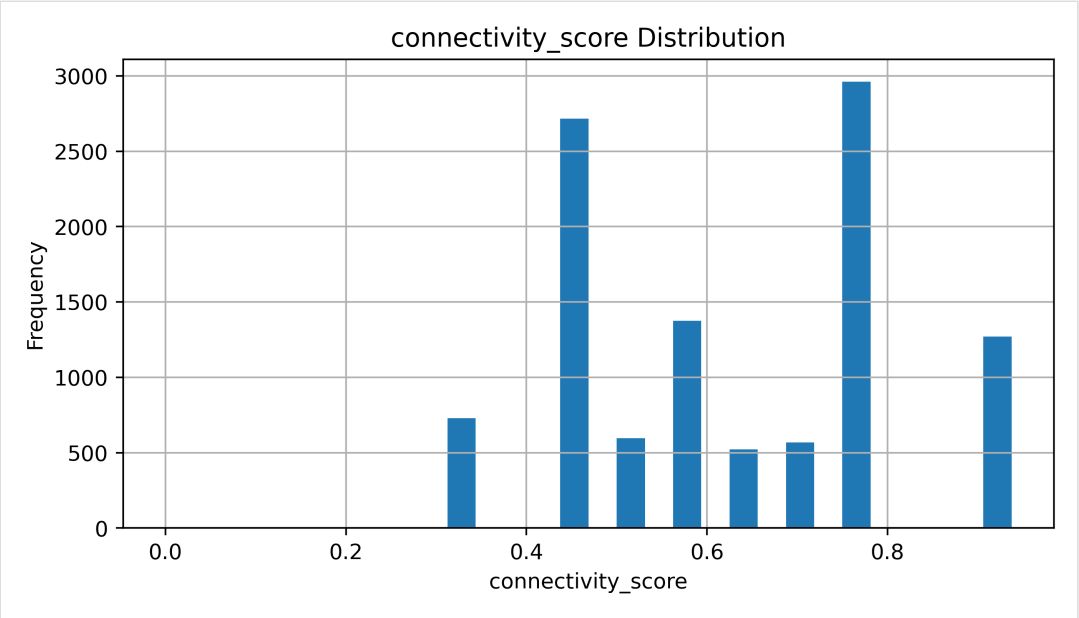
Histogram of lane readiness score. Higher values indicate stronger lane detection confidence/quality.



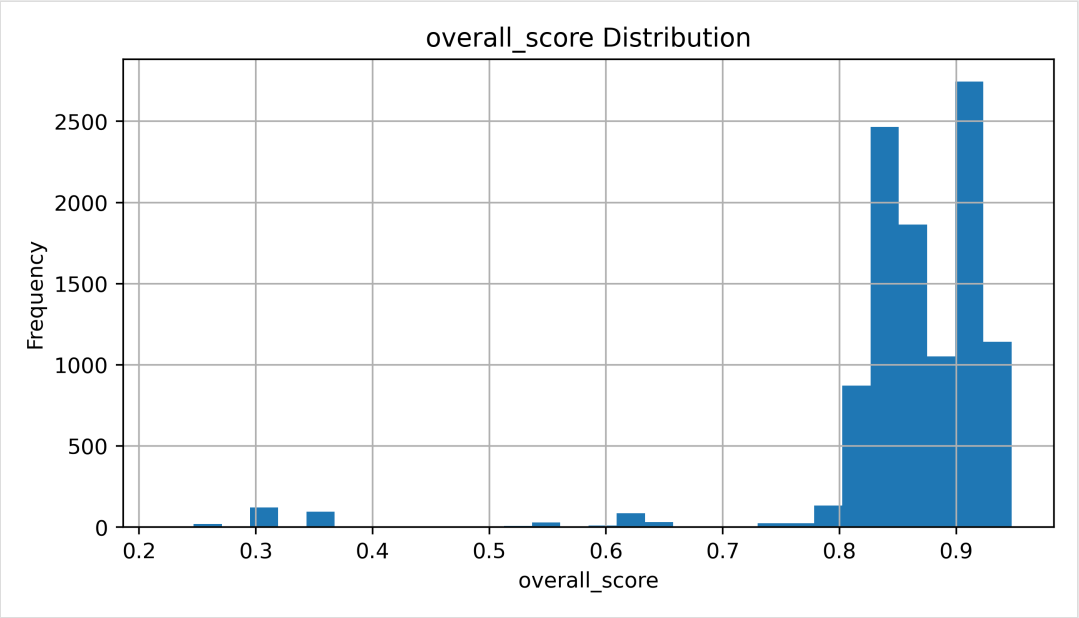
Histogram of GPS quality score derived from localization error.



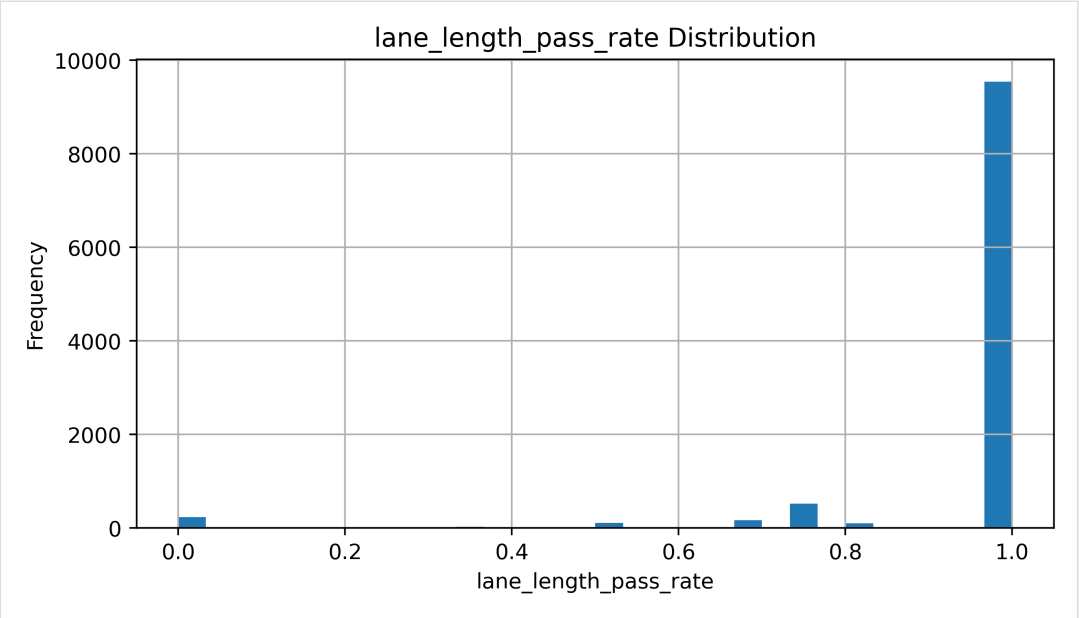
Histogram of connectivity score from throughput/latency metrics.



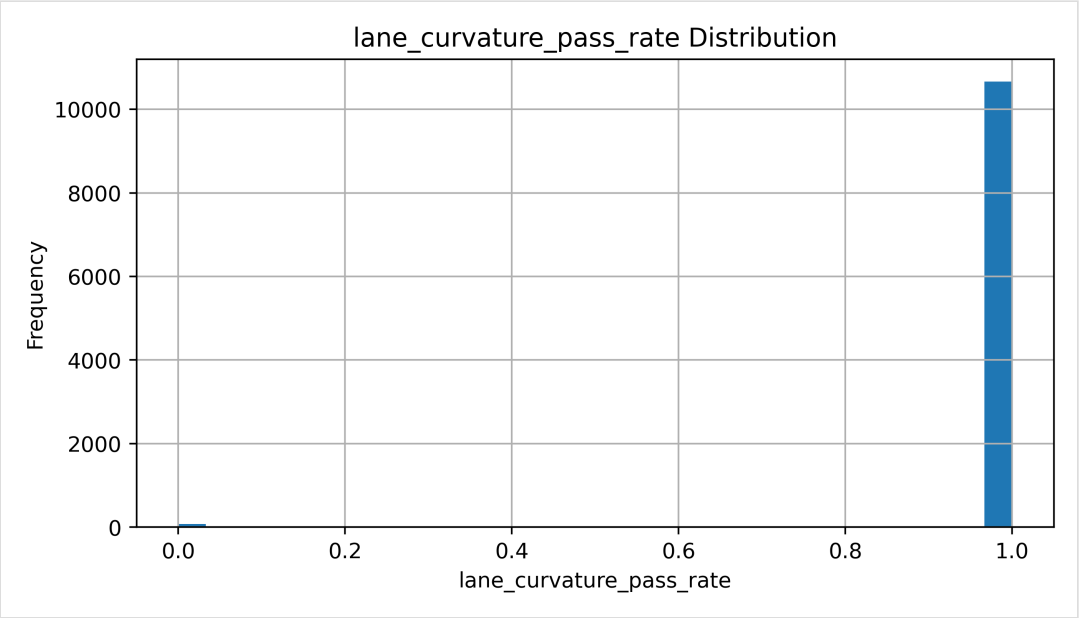
Histogram of final combined readiness score.



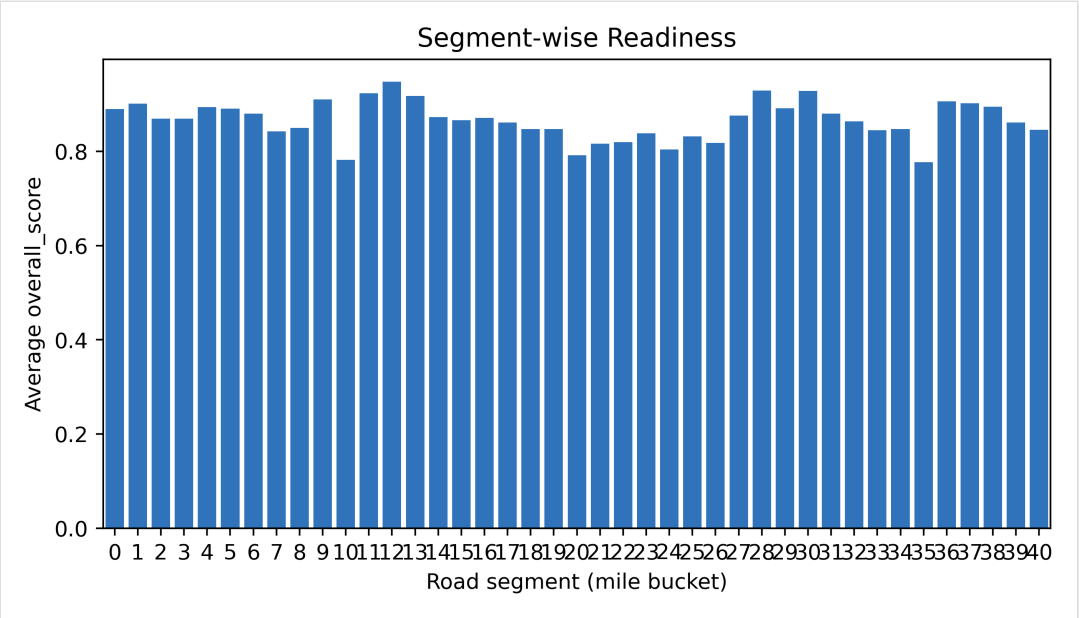
Distribution of pass-rate for lane minimum-length checks per frame.



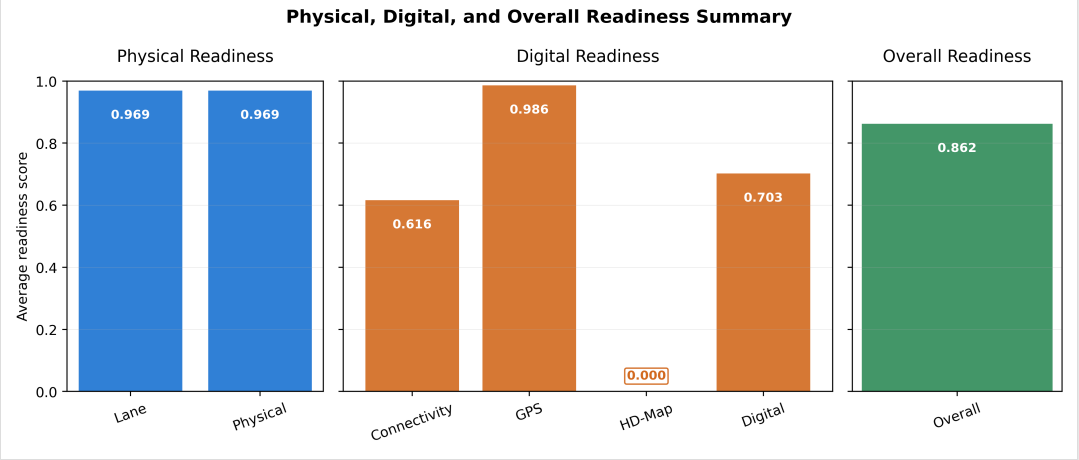
Distribution of pass-rate for lane curvature checks per frame.



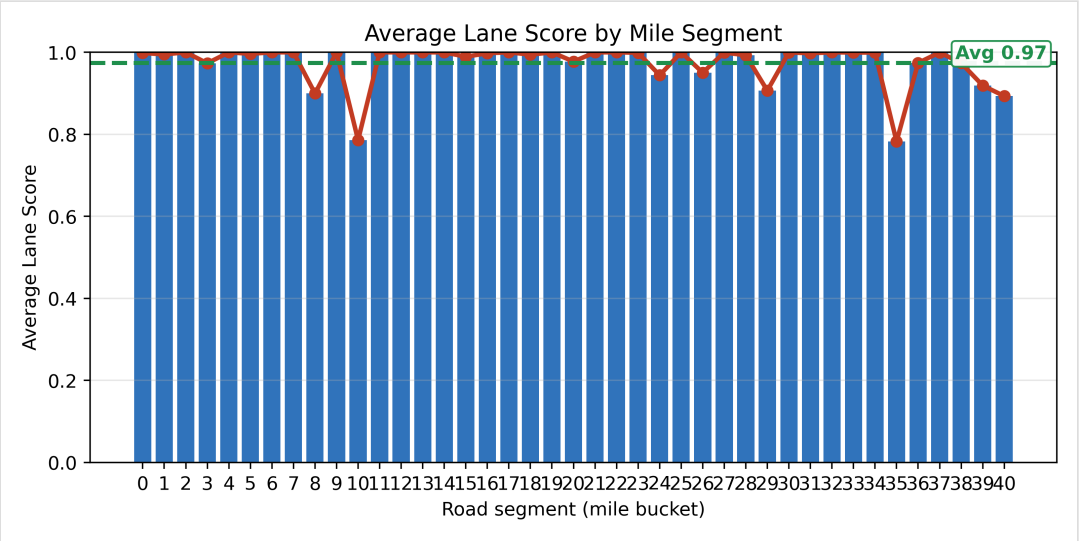
Average overall readiness by road segment (mile bucket), useful for hotspot identification.



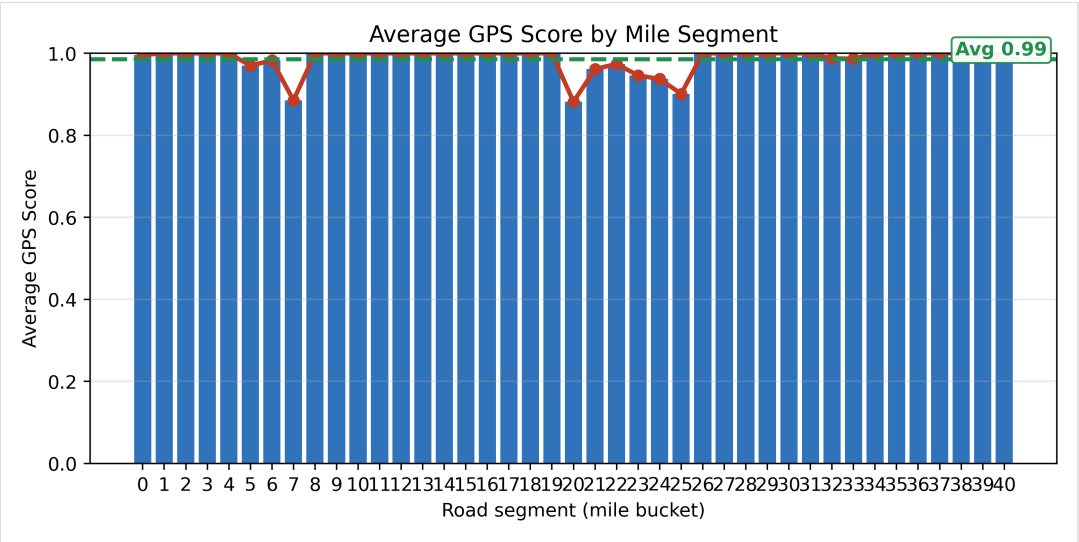
Summary chart comparing average physical readiness, digital readiness, and overall readiness, along with their component scores.



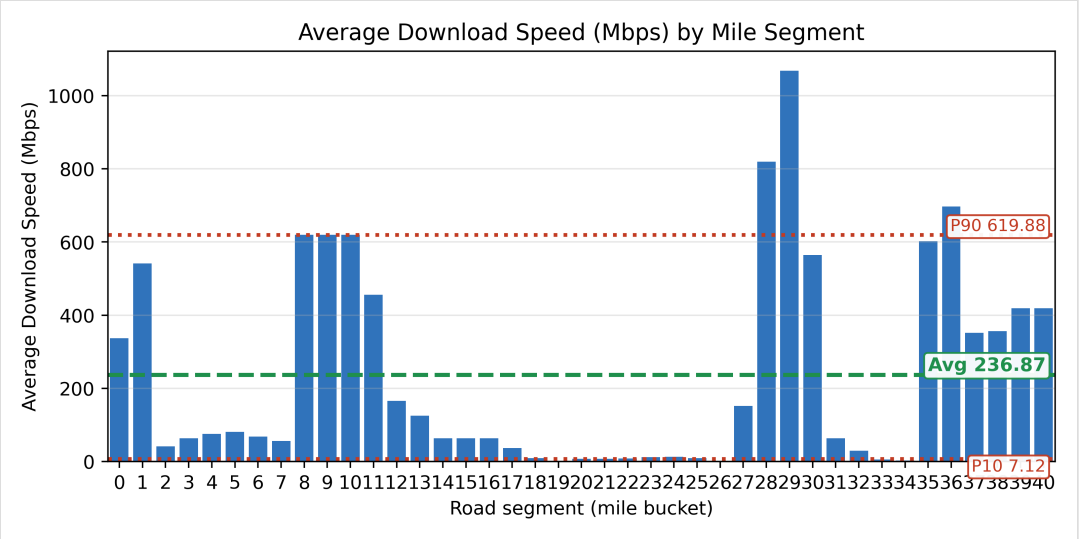
Bar chart of average lane readiness score per mile segment with average, P10, and P90 reference lines.



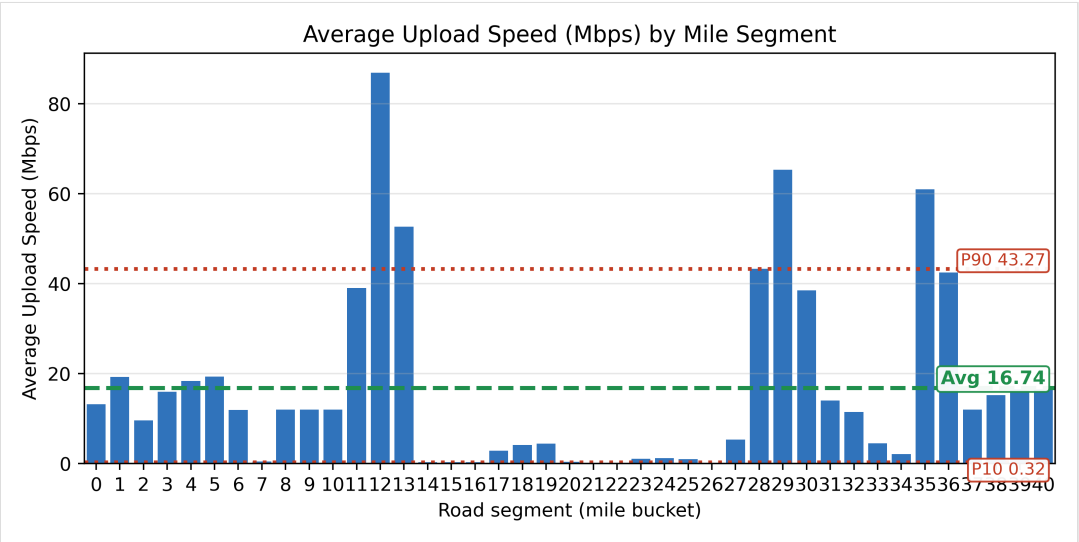
Bar chart of average GPS readiness score per mile segment with average, P10, and P90 reference lines.



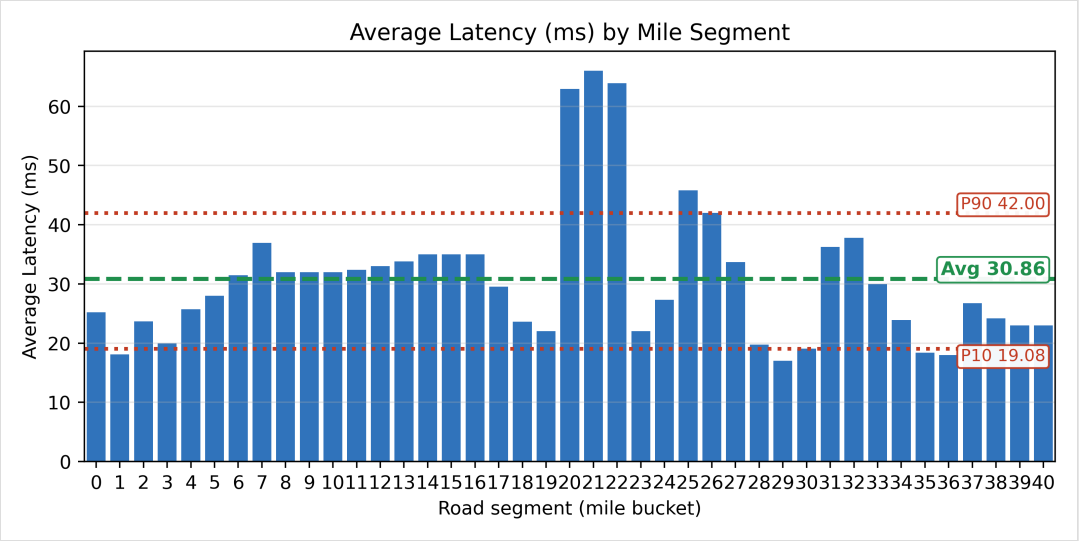
Bar chart of average download speed (Mbps) per mile segment with average, P10, and P90 reference lines.



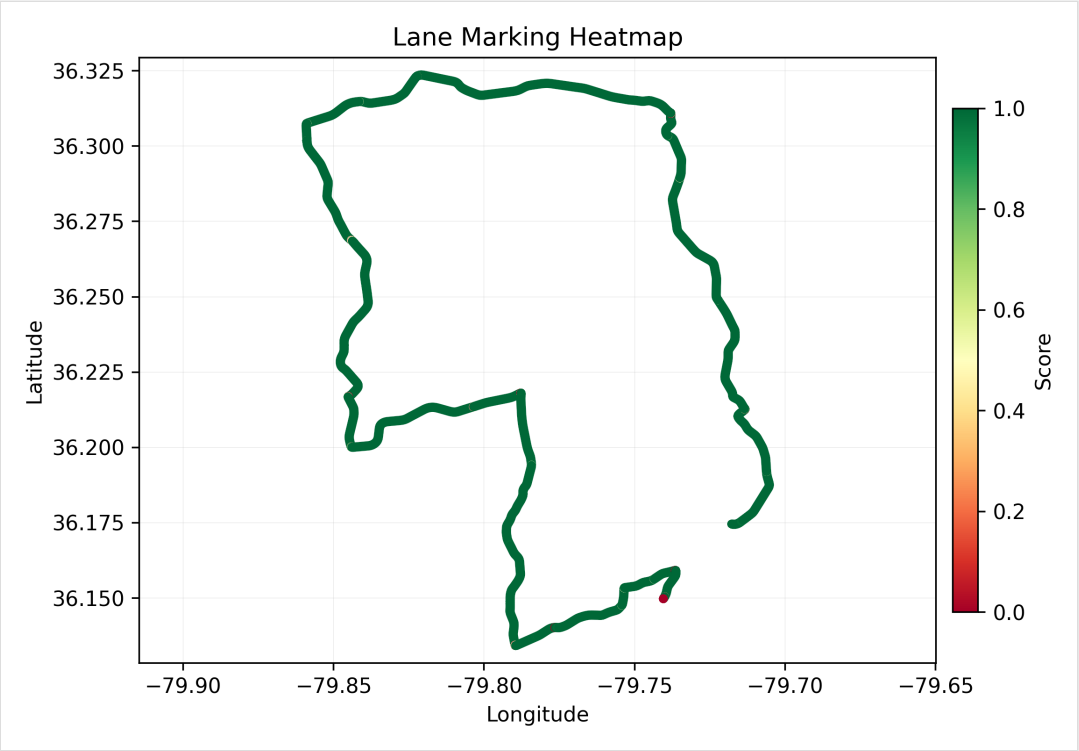
Bar chart of average upload speed (Mbps) per mile segment with average, P10, and P90 reference lines.



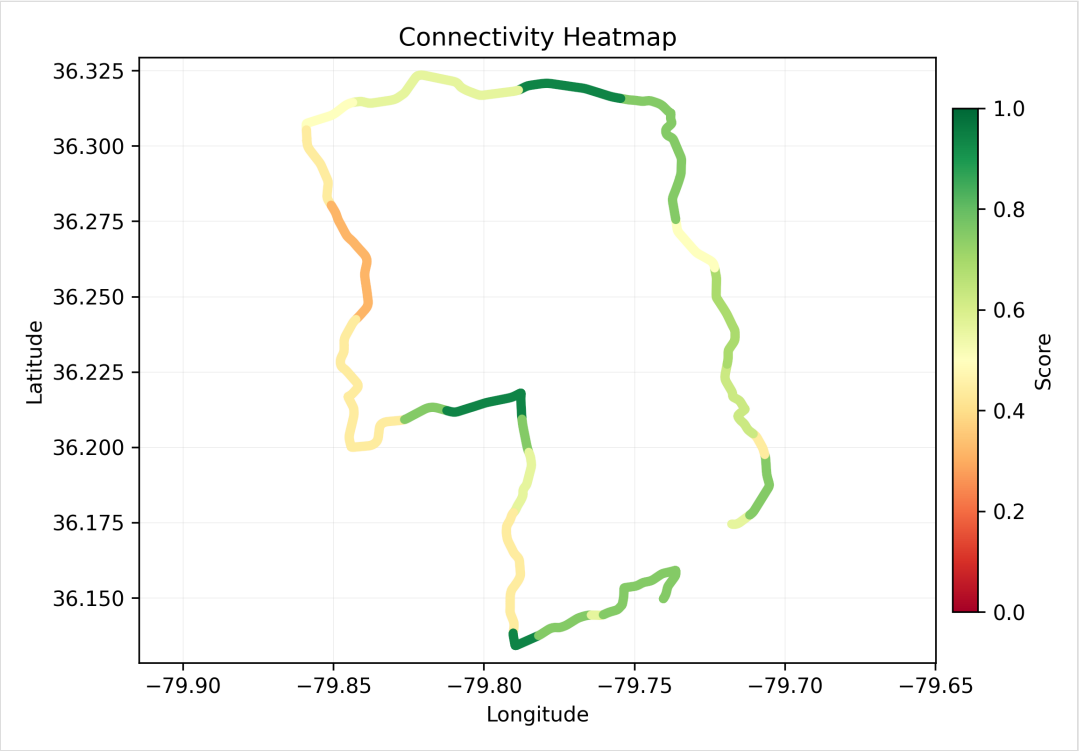
Bar chart of average latency (ms) per mile segment with average, P10, and P90 reference lines. Ookla provides a single latency measure, so upload/download latency are not shown separately.



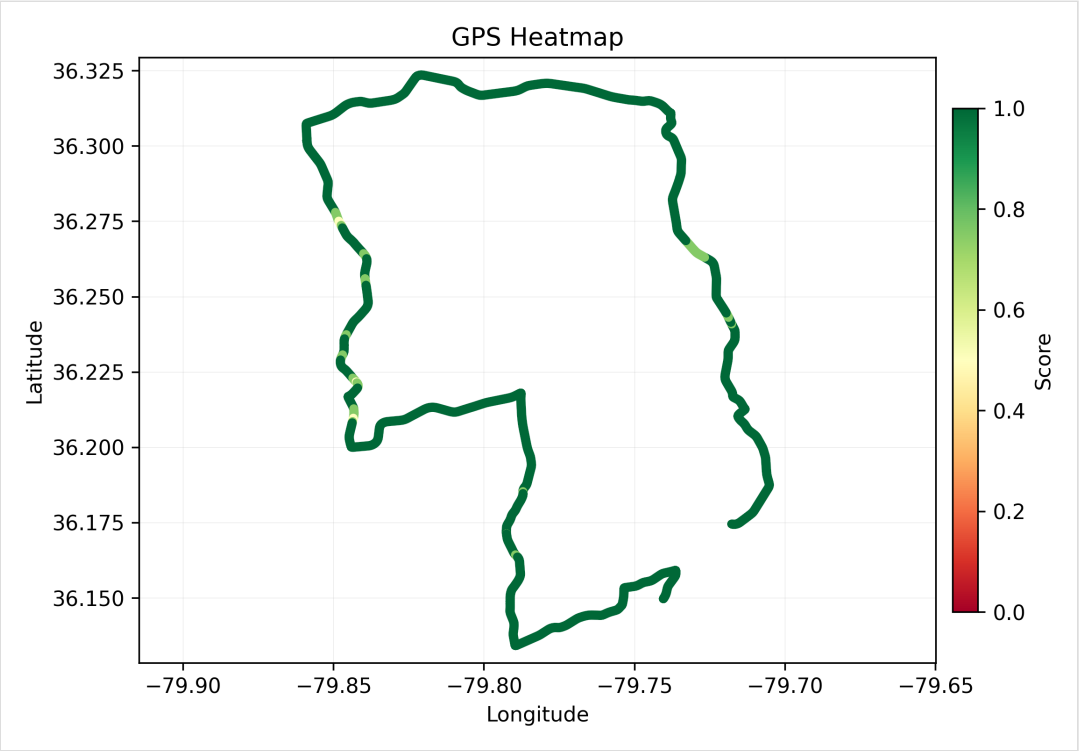
Route heatmap for the lane-marking component score.



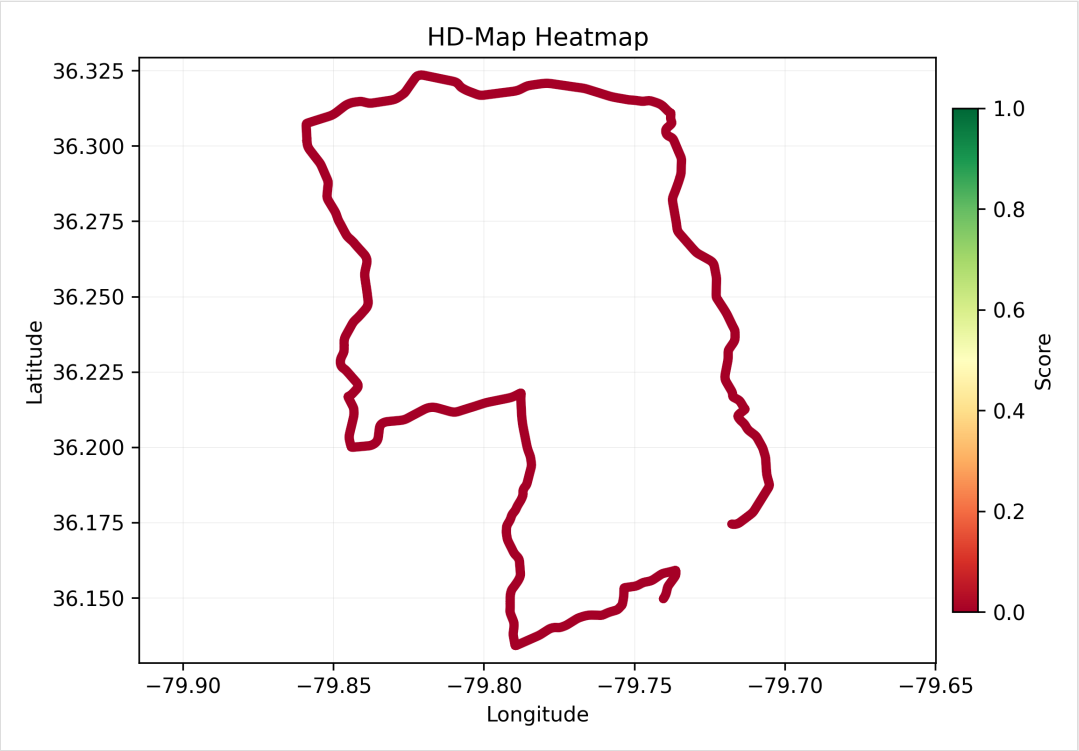
Route heatmap for the connectivity component score.



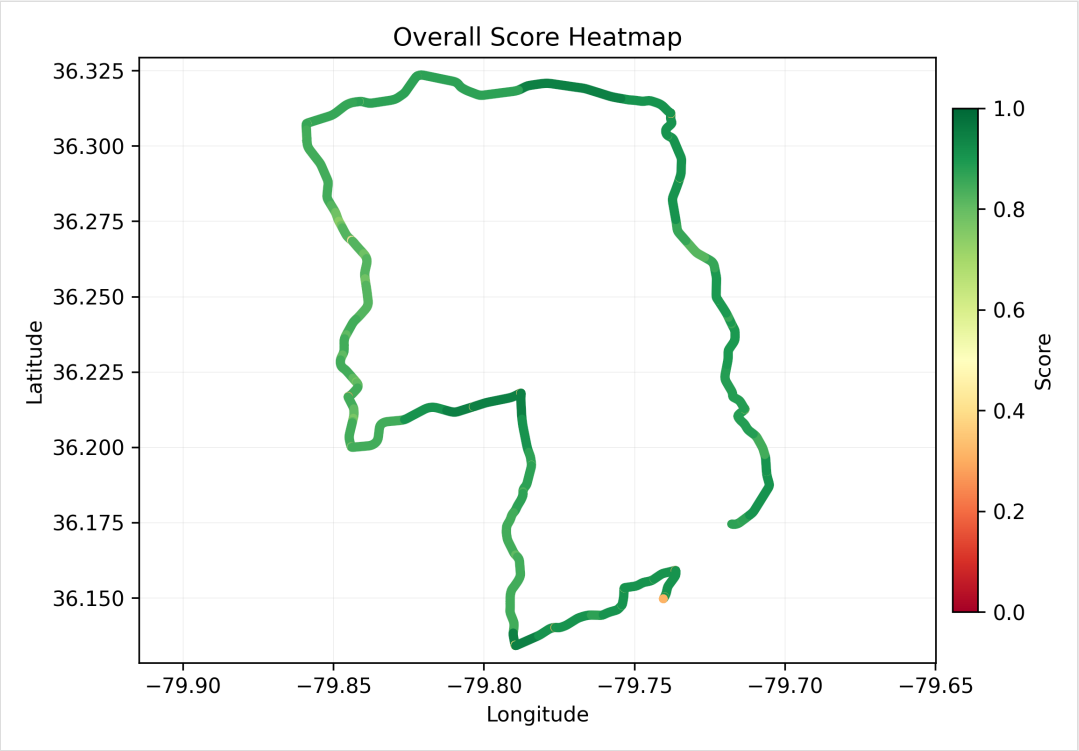
Route heatmap for the GPS component score.



Route heatmap for the HD-map component score.



Route heatmap for the overall readiness score.



Issue Evidence Images

Segment 35 | frame 93280 | overall=0.348, lane=0.000, gps=1.000, conn=0.938, lane Leng/Curv/Sepa=(0.00/1.00/0.00) | issues: `lane quality low`, `lane length check weak`, `hd-map coverage low`, `overall readiness low`

Segment 35 | frame 93720 | overall=0.348, lane=0.000, gps=1.000, conn=0.938, lane Leng/Curv/Sepa=(0.00/1.00/0.00) | issues: `lane quality low`, `lane length check weak`, `hd-map coverage low`, `overall readiness low`

Segment 10 | frame 27650 | overall=0.310, lane=0.000, gps=1.000, conn=0.750, lane Leng/Curv/Sepa=(0.00/1.00/0.00) | issues: `lane quality low`, `lane length check weak`, `hd-map coverage low`, `overall readiness low`

Segment 10 | frame 28120 | overall=0.310, lane=0.000, gps=1.000, conn=0.750, lane Leng/Curv/Sepa=(0.00/0.00/0.00) | issues: `lane quality low`, `hd-map coverage low`, `overall readiness low`

Segment 20 | frame 54370 | overall=0.223, lane=0.000, gps=1.000, conn=0.312, lane Leng/Curv/Sepa=(0.00/1.00/0.00) | issues: `lane quality low`, `lane length check weak`, `connectivity low`, `hd-map coverage low`, `overall readiness low`

Segment 20 | frame 54380 | overall=0.223, lane=0.000, gps=1.000, conn=0.312, lane Leng/Curv/Sepa=(0.00/0.00/0.00) | issues: `lane quality low`, `connectivity low`, `hd-map coverage low`, `overall readiness low`

Evaluation Data Snapshot Sample (Top 40)

frame_index	image	segment_id	lane_score	gps_score	connectivity_score	overall_score
0	image_0.jpg	0	1	0	0	0.6
10	image_10.jpg	0	1	1	0.5625	0.8725
20	image_20.jpg	0	0.99	1	0.5625	0.8665
30	image_30.jpg	0	1	1	0.5625	0.8725
40	image_40.jpg	0	1	1	0.5625	0.8725
50	image_50.jpg	0	1	1	0.5625	0.8725
60	image_60.jpg	0	1	1	0.5625	0.8725
70	image_70.jpg	0	1	1	0.5625	0.8725
80	image_80.jpg	0	1	1	0.5625	0.8725
90	image_90.jpg	0	1	1	0.5625	0.8725
100	image_100.jpg	0	1	1	0.5625	0.8725
110	image_110.jpg	0	1	1	0.5625	0.8725
120	image_120.jpg	0	1	1	0.5625	0.8725
130	image_130.jpg	0	1	1	0.5625	0.8725
140	image_140.jpg	0	1	1	0.5625	0.8725
150	image_150.jpg	0	1	1	0.5625	0.8725
160	image_160.jpg	0	1	1	0.5625	0.8725
170	image_170.jpg	0	1	1	0.5625	0.8725
180	image_180.jpg	0	1	1	0.5625	0.8725
190	image_190.jpg	0	1	1	0.5625	0.8725
200	image_200.jpg	0	1	1	0.5625	0.8725
210	image_210.jpg	0	1	1	0.5625	0.8725

frame_index	image	segment_id	lane_score	gps_score	connectivity_score	overall_score
220	image_220.jpg	0	1	1	0.5625	0.8725
230	image_230.jpg	0	1	1	0.5625	0.8725
240	image_240.jpg	0	1	1	0.5625	0.8725
250	image_250.jpg	0	1	1	0.5625	0.8725
260	image_260.jpg	0	1	1	0.5625	0.8725
270	image_270.jpg	0	1	1	0.5625	0.8725
280	image_280.jpg	0	1	1	0.5625	0.8725
290	image_290.jpg	0	1	1	0.5625	0.8725
300	image_300.jpg	0	1	1	0.5625	0.8725
310	image_310.jpg	0	1	1	0.5625	0.8725
320	image_320.jpg	0	1	1	0.5625	0.8725
330	image_330.jpg	0	1	1	0.5625	0.8725
340	image_340.jpg	0	1	1	0.5625	0.8725
350	image_350.jpg	0	1	1	0.5625	0.8725
360	image_360.jpg	0	1	1	0.5625	0.8725
370	image_370.jpg	0	1	1	0.5625	0.8725
380	image_380.jpg	0	1	1	0.5625	0.8725
390	image_390.jpg	0	1	1	0.5625	0.8725